

Basic Statistics

BCA — Semester 2 — 2020 — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What is the difference between relative and absolute statistical measure? J.J. Thomson discovered the electron by isolating negatively charged particles for which he could measure the mass-charge ratio. This ratio appeared to be Constant over a wide range of experimental conditions and, consequently, could be characteristics of new particles. His observations; from two different Cathode- ray tubes that used air as the gas. [10]Tube I 0.570.340.430.320.480.400.40Tube II 0.530.470.470.510.630.610.48 Which tube is more homogeneous?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. In the Nepal, during the fall harvest season Pumpkin are sold in the large quantities at the farm stands. Often instead of weighting the Pumpkin prior to sale, the farm stand operators will just places the Pumpkin in the appropriate circular cutout on the counter. When asked why this was done one farmer replied. can tell the weight of the Pumpkin from its circumference". To determine whether this was really true; a sample of 8 Pumpkin's circumference and weights were measured: [10]Circumference (cm) 50.55 54.52 37.52 53.47 Weight (gram) 1220 1517 5010 1514a. Develop the estimating linear regression equation of weight of the Pumpkin to the circumference, and b. Estimate the weight of Pumpkin if the circumference value is 40 cm. Can it be possible to predict the weight of a pumpkin if it's circumference if 80 cm?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define normal distribution and standard normal distribution. In an examination, 10% students got less than 20 marks and 95% got less than 75 marks. Assuming the distribution to be normal, find the mean and standard deviation. [10]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What do you mean by secondary data? Write the source of secondary data with special focus to Nepalese context. [5]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Construct histogram, frequency polygon and frequency curve of the following frequency distribution. [5]
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Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Following data represent the preference of 10 students studying BIT towards two brands of computers namely DELL and HP. [5]
Computer Student preference DELL 52981103467 HP 10513862794
Apply Pearson's rank correlation coefficient and compute correlation coefficient. Also interpret your result.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Define random variable. A random variable has the following probability function. [5]
X-2-1012P(X) 0.2 K 0.42 K K Find (i) K, (ii) $P(X \geq 0)$, (iii) $E(X)$, and (iv) $E(4X+5)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. During one stage in the manufacture of integrated circuit chips, a coating must be applied. If 70% of chips received a thick enough coating, find the probability that among 15 chips (i) at least 12 will have thick enough coatings, (ii) at most 2 will have thick enough coatings, and (iii) exactly 10 will have thick enough coatings. [5]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. A new computer virus can enter the system through e-mail or through the internet. There is a 30% chance of receiving this virus through e-mail. There is a 40% chance of receiving it through the internet. Also, the virus enters the system simultaneously through e-mail and the internet with probability 0.15. What is, the probability that the virus does not enter the system at all? [5]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. The standard deviation of a symmetric distribution is 9. Compute the possible value of fourth central moment for the distribution to be (i) mesokurtic (ii) platykurtic, and (iii) leptokurtic. [5]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. The mass of vitamin E in a capsule manufactured by a certain drug company is normally distributed with standard deviation 0.042 mg. A random sample of 36 capsules was analyzed and the mean mass of vitamin E was found to be 5.12 mg. Calculate a symmetric 95% confidence interval for the population mean mass of vitamin E per capsule. [5]

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on the following: [2.5x2=5]
a. Box and whisker plot. b. Census survey and sample survey.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.